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Analysis of GAR and NAR using NSSO Education Survey Data

A PROJECT REPORT

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Abstract

This study uses unit-level data from the NSSO 75th Round (2017-18) Social Consumption: Education Survey to examine educational attendance in India. The Gross Attendance Ratio (GAR) and Net Attendance Ratio (NAR) are estimated for four educational levels Primary, Upper Primary/Middle, Secondary, and Higher Secondary. The analysis provides All-India estimates as well as estimates for selected states, namely Uttar Pradesh (state code 09), Tamil Nadu (state code 33) and D & N Haveli (state code 26). These states were selected to represent different socio-economic and demographic contexts: Uttar Pradesh as a large and populous state with relatively lower educational indicators, Tamil Nadu as a state with comparatively better educational outcomes, and D & N Haveli as a small Union Territory to capture variations in smaller administrative regions. Estimates are further disaggregated by gender and rural-urban sector, and by UMPCE quintile classes to examine socio-economic disparities. Relative Standard Errors (RSE) are computed to assess the reliability of the estimates.

Introduction

Education plays a vital role in improving human capital and promoting socio-economic development. Measuring school participation helps in assessing access to education and identifying inequalities across different population groups. In India, the National Sample Survey Office (NSSO) conducts periodic large-scale household surveys that provide valuable data on educational attendance and related characteristics.

Two important indicators used to measure educational participation are the Gross Attendance Ratio (GAR) and the Net Attendance Ratio (NAR). GAR measures the proportion of individuals attending a particular level of education relative to the population in the corresponding age group, regardless of age. In contrast, NAR measures attendance only among individuals who belong to the official age group for that level of education. While GAR indicates the overall participation in a level of education, NAR reflects age-appropriate attendance.

This study uses unit-level data from the NSS 75th Round Education Survey (2017-18) to estimate GAR and NAR for different educational levels. In addition to All-India estimates, the analysis focuses on three selected regions, Uttar Pradesh, Tamil Nadu and D & N Haveli. These regions were chosen to represent diversity in India's educational landscape. Uttar Pradesh is one of the most populous states with significant rural population and varying educational outcomes, Tamil Nadu is known for relatively strong educational indicators and higher literacy rates, and D & N Haveli represents a smaller administrative region with different demographic and socio-economic characteristics.

The estimates are computed separately for rural and urban sectors and for males and females. Further analysis across UMPCE quintile classes is carried out to examine socio-economic differences in educational attendance. To evaluate the statistical reliability of the results, Relative Standard Errors (RSE) are also calculated for each estimate.

Problem Statement

The objective of this study is to analyze educational attendance patterns in India using the unit-level micro-data from the NSSO 75th Round (2017-18) Social Consumption: Education Survey. The study focuses on estimating the Gross Attendance Ratio (GAR) and Net Attendance Ratio (NAR) for different levels of education and examining variations across demographic and socio-economic groups.

The specific objectives of the study are as follows:

1. To estimate the Gross Attendance Ratio (GAR) and Net Attendance Ratio (NAR) for the following educational levels:

- Primary
- Upper Primary / Middle
- Secondary
- Higher Secondary

The estimates are obtained for both All-India and selected states, namely Uttar Pradesh, Tamil Nadu and D & N Haveli, separately for rural and urban sectors and for males and females.

2. To compute the Relative Standard Error (RSE) for each estimate of GAR and NAR in order to evaluate the statistical reliability of the estimates.
3. To examine socio-economic disparities in educational attendance by estimating GAR and NAR across UMPCE quintile classes, separately for rural and urban sectors.
4. To obtain age-specific estimates of GAR and NAR corresponding to the relevant age groups for different educational levels, and to compare the gap between GAR and NAR across gender and sector.

Through these analyses, the study aims to provide insights into educational participation across regions, gender groups, sectors, and socio-economic classes in India.

Sampling Design and Estimation Procedure

Definitions

The analysis of educational attendance in this study is based on three indicators: Gross Attendance Ratio (GAR), Net Attendance Ratio (NAR), and Relative Standard Error (RSE).

Gross Attendance Ratio (GAR) measures the overall participation in a particular level of education. It is defined as the ratio of the number of persons attending a given level of education to the estimated population in the official age group corresponding to that level. For example, for primary education,

$$GAR_{primary} = \frac{\text{Number of persons attending Primary}}{\text{Estimated population aged 6-10 years}} \times 100$$

GAR may exceed 100 percent because it includes students who are younger or older than the official age group for that level of education.

Net Attendance Ratio (NAR) measures age-appropriate participation in education. It is defined as the ratio of the number of persons in the official age group attending the corresponding level of education to the total number of persons in that age group.

$$NAR_{primary} = \frac{\text{Persons aged 6-10 attending Primary}}{\text{Estimated population aged 6-10}} \times 100$$

Unlike GAR, NAR cannot exceed 100 percent because only individuals within the relevant age group are included.

Following is the official age groups corresponding to different levels of education used for computing GAR and NAR,

Education Level	Classes Covered	Official Age Group (years)
Primary	Classes I-V	6-10
Upper Primary / Middle	Classes VI-VIII	11-13
Secondary	Classes IX-X	14-15
Higher Secondary	Classes XI-XII	16-17

Sampling Design

The NSS 75th Round Education Survey follows a **multi-stage stratified sampling design**. The survey covers the entire country and produces estimates for All-India, States/UTs, and rural-urban sectors separately.

In the first stage, the population is divided into **strata and sub-strata** separately for rural and urban sectors. From each sub-stratum, **First Stage Units (FSUs)** are selected. In rural areas, villages serve as FSUs, while in urban areas, **Urban Frame Survey (UFS) blocks** are used as FSUs. FSUs are generally selected using **Probability Proportional to Size with Replacement (PPSWR)** where the size is measured in terms of population or number of households.

In the second stage, households are selected from the sampled FSUs. Large FSUs are first divided into **hamlet-groups (in rural areas)** or **sub-blocks (in urban areas)**. From these, selected groups are listed and households are classified into **Second Stage Strata (SSS)** based on educational characteristics of household members. Finally, households are selected from each SSS using **Simple Random Sampling Without Replacement (SRSWOR)**.

This multi-stage procedure ensures representativeness of the sample and allows unbiased estimation of population parameters.

Estimation Procedure

Let s denote the stratum, t the sub-stratum, m the sub-sample ($m = 1, 2$), i the First Stage Unit (FSU), j the second-stage stratum, and k the household. Let Z denote the total size of the sub-stratum and z_i the size of the sampled FSU. Further, H and h represent the number of listed and surveyed households respectively, and y and x denote the characteristics under estimation.

The estimate of the aggregate for the j -th second-stage stratum is given by

$$\hat{Y}_j = \frac{Z}{n_j} \sum_{i=1}^{n_j} \frac{1}{z_i} \left[\frac{H_{i1j}}{h_{i1j}} \sum_{k=1}^{h_{i1j}} y_{i1jk} + D_i^* \frac{H_{i2j}}{h_{i2j}} \sum_{k=1}^{h_{i2j}} y_{i2jk} \right]$$

The estimate for a sub-stratum is obtained by averaging the estimates from the two sub-samples is:

$$\hat{Y}_{st} = \frac{1}{2} \sum_{m=1}^2 \hat{Y}_{stm}$$

and the estimate for a stratum is:

$$\hat{Y}_s = \sum_t \hat{Y}_{st}$$

and the overall estimate (State/UT or All-India) is obtained as

$$\hat{Y} = \sum_s \hat{Y}_s$$

For ratio-type indicators such as GAR and NAR, the **combined ratio estimator** is used. If $\hat{Y}$ and $\hat{X}$ are the estimated aggregates of the numerator and denominator respectively, then the ratio estimate is

$$\hat{R} = \frac{\hat{Y}}{\hat{X}}$$

The mean squared error of the ratio estimator is given by

$$M\hat{S}E(\hat{R}) = \frac{1}{4\hat{X}^2} \sum_s \sum_t \left[(\hat{Y}_{st1} - \hat{Y}_{st2})^2 + \hat{R}^2 (\hat{X}_{st1} - \hat{X}_{st2})^2 - 2\hat{R}(\hat{Y}_{st1} - \hat{Y}_{st2})(\hat{X}_{st1} - \hat{X}_{st2}) \right]$$

and RSE is defined as,

$$RSE(\hat{R}) = \frac{\sqrt{M\hat{SE}(\hat{R})}}{\hat{R}} \times 100$$

In the present analysis, the survey multiplier `mult_combined` is used to obtain estimates of aggregates Y and X , while `mult_subsample` is used to obtain sub-sample level estimates required for variance and RSE computation.

Multiplier Concept

In the NSS 75th Round, survey multipliers (weights) are used to expand sample observations to obtain population level estimates. The multiplier provided in the NSS documentation corresponds to the **sub-sample multiplier** used for variance estimation.

The multiplier is given by

$$w = \frac{Z_{st}}{n_{st}} \times \frac{1}{z_{stmi}} \times \frac{H_{stmi1j}}{h_{stmi1j}} \quad (\text{for hg/sb} = 1)$$

$$w = \frac{Z_{st}}{n_{st}} \times \frac{1}{z_{stmi}} \times D_{stmi}^* \times \frac{H_{stmi2j}}{h_{stmi2j}} \quad (\text{for hg/sb} \geq 2)$$

In the unit level dataset, two multiplier variables are provided:

- `mult_subsample`: multiplier corresponding to each sub-sample used for variance estimation.
- `mult_combined`: multiplier used for computing overall population estimates.

The combined multiplier is obtained by averaging the two paired sub-samples. Therefore,

$$w_{\text{combined}} = \frac{1}{2} w_{\text{subsample}}$$

Using the combined multiplier, the estimate of a population aggregate is

$$\hat{Y} = \sum_i w_i^{(\text{combined})} y_i$$

For variance estimation, aggregates are computed separately for each sub-sample:

$$\hat{Y}_{stm} = \sum_i w_i^{(\text{subsample})} y_i, \quad m = 1, 2$$

These sub-sample estimates are then used to compute the variance and Relative Standard Error (RSE) using the paired sub-sample variance formulas.

Outputs and Observations

Estimates of GAR by Gender & Sector

Table 1: Gross Attendance Ratio (GAR) by Sector, Course and Gender

Sector	Course	Gender	All India	UP (9)	D & N Haveli (26)	Tamil Nadu (33)
Rural	Primary	Male	101.70	101.78	100.00	98.84
		Female	99.89	95.30	89.54	99.88
	Upper Primary / Middle	Male	94.80	87.88	101.94	95.74
		Female	94.16	93.45	104.60	91.68
	Secondary	Male	85.29	68.65	84.81	89.66
		Female	82.34	54.48	63.48	98.19
	Higher Secondary	Male	66.38	63.24	31.45	81.04
		Female	61.13	53.92	126.89	98.81
Urban	Primary	Male	102.40	105.31	94.39	99.75
		Female	101.97	92.32	100.00	121.53
	Upper Primary / Middle	Male	94.26	78.29	76.60	104.60
		Female	93.76	87.37	83.21	93.78
	Secondary	Male	93.78	73.11	119.86	85.87
		Female	93.67	72.61	142.97	83.08
	Higher Secondary	Male	80.15	72.61	73.24	90.08
		Female	79.16	61.14	66.52	105.19

Key Observations on Gross Attendance Ratio (GAR)

- **Universal Primary Access:** GAR values at the Primary level are consistently near or above 100 across all observed states. For instance, All-India Rural Male GAR stands at 101.70, indicating high levels of initial enrollment.
- **Educational Attrition:** There is a significant decline in attendance as the level of education increases. The All-India Rural GAR drops from approximately 101% at the Primary level to roughly 61-66% at the Higher Secondary level.
- **Regional Excellence (Tamil Nadu):** Tamil Nadu (State 33) exhibits superior retention rates. Its Rural Female GAR for Higher Secondary (98.81) far exceeds the National average (61.13) and Uttar Pradesh (53.92).
- **Urban Advantage:** Urban areas consistently report higher GARs for secondary education. The All-India Higher Secondary GAR for Urban Females (79.16) is eighteen percentage points higher than its Rural counterpart (61.14).
- **Gender Disparity in Uttar Pradesh:** A prominent gender gap is observed in rural Uttar Pradesh (State 9) at the Secondary level, where the male GAR (68.65) is significantly higher than the female GAR (54.48).
- **Statistical Volatility:** Data for D & N Haveli (State 26) shows extreme values (e.g., 126.89 for Rural Higher Secondary Females), which, when paired with high Relative Standard Error (RSE) values, indicates potential outliers due to small sample sizes.

Estimates of NAR by Gender & Sector

Key Observations on Net Attendance Ratio (NAR)

- **Lower Baseline vs. GAR:** NAR values are consistently lower than GAR, reflecting that a significant portion of the student population is not in the age-appropriate grade. National primary NAR sits around 85-87%, whereas GAR was near 100%.

Table 2: Net Attendance Ratio (NAR) by Sector, Course, and Gender

Sector	Course	Gender	All India	UP (9)	D & N Haveli (26)	Tamil Nadu (33)
Rural	Primary	Male	86.56	80.98	100.00	88.61
		Female	84.81	78.25	89.54	88.72
	Upper Primary / Middle	Male	72.12	58.19	97.20	81.02
		Female	70.75	61.82	95.66	75.41
	Secondary	Male	56.67	41.00	79.35	67.84
		Female	55.18	33.89	58.22	69.06
	Higher Secondary	Male	40.29	30.90	30.45	61.21
		Female	39.24	31.77	21.13	63.18
Urban	Primary	Male	87.67	79.18	74.73	87.50
		Female	86.22	73.89	100.00	95.58
	Upper Primary / Middle	Male	73.48	55.84	76.60	81.01
		Female	74.98	62.89	77.23	89.66
	Secondary	Male	61.52	39.45	79.31	62.61
		Female	63.65	47.82	88.30	66.59
	Higher Secondary	Male	53.11	39.61	11.75	55.92
		Female	52.35	40.93	65.74	67.63

- **Significant Attrition:** There is a sharp decline in age-appropriate attendance. The All-India Rural NAR drops from 86.56% at the Primary level to 40.29% at the Higher Secondary level, representing a 53% reduction in participation.
- **Regional Performance (Tamil Nadu):** Tamil Nadu (State 33) shows the highest age-appropriate retention. Its Higher Secondary NAR (approx. 56-67%) is significantly higher than the National average (approx. 40-53%).
- **Urban-Rural Disparity:** The attendance gap between sectors increases with the level of education. At the Higher Secondary level, Urban NAR (53.11%) is over 12 percentage points higher than Rural NAR (40.29%) for males.
- **Gender Gap in Uttar Pradesh:** Rural Uttar Pradesh (State 9) shows a distinct gender disadvantage at the Secondary level, with a Female NAR of 33.89% compared to 41.00% for Males.
- **Data Volatility:** In D & N Haveli (State 26), extremely low NAR values at higher levels (e.g., 21.13% for Rural Females) paired with high RSE (57.20%) suggest limited sample sizes for those specific cohorts.

Estimates of Relative Standard Error (RSE)

Key Observations on Statistical Reliability (RSE)

- **National Level Robustness:** All-India RSE values for both GAR and NAR remain consistently below 4%, indicating a high degree of confidence in the national aggregate figures.

Table 3: Gross Attendance Ratio Relative Standard Error (GAR_RSE) by Sector, Course and Gender

Sector	Course	Gender	All India	UP (9)	D & N Haveli (26)	Tamil Nadu (33)
Rural	Primary	Male	0.70	1.86	0.00	3.35
		Female	0.87	2.21	11.53	3.70
	Upper Primary / Middle	Male	1.25	3.88	4.52	5.15
		Female	1.53	4.32	9.40	5.75
	Secondary	Male	1.95	5.66	25.52	8.06
		Female	1.96	7.03	39.39	8.83
	Higher Secondary	Male	2.64	7.16	65.17	8.28
		Female	3.02	7.89	92.39	8.79
Urban	Primary	Male	0.94	3.02	20.65	4.45
		Female	1.17	3.89	0.00	4.56
	Upper Primary / Middle	Male	1.77	6.27	7.17	5.85
		Female	1.98	6.71	32.97	2.83
	Secondary	Male	2.48	7.28	36.58	9.85
		Female	2.40	7.35	36.15	8.47
	Higher Secondary	Male	2.89	10.54	108.34	10.54
		Female	3.03	9.68	31.35	9.65

Table 4: Net Attendance Ratio Relative Standard Error (NAR_RSE) by Sector, Course and Gender

Sector	Course	Gender	All India	UP (9)	D & N Haveli (26)	Tamil Nadu (33)
Rural	Primary	Male	0.58	1.62	0.00	2.49
		Female	0.68	2.00	11.53	2.56
	Upper Primary / Middle	Male	1.09	4.30	3.87	4.70
		Female	1.33	3.67	3.81	4.87
	Secondary	Male	1.86	6.14	23.34	6.62
		Female	1.99	8.23	44.41	7.37
	Higher Secondary	Male	2.79	8.39	71.22	8.53
		Female	3.07	7.66	57.20	8.27
Urban	Primary	Male	0.74	2.28	22.47	3.31
		Female	0.90	3.73	0.00	1.81
	Upper Primary / Middle	Male	1.42	6.20	7.17	4.69
		Female	1.67	6.60	28.26	2.54
	Secondary	Male	2.26	8.32	24.10	10.93
		Female	2.21	7.42	15.60	8.40
	Higher Secondary	Male	3.55	10.85	39.72	7.45
		Female	2.98	9.40	31.85	6.75

- **Inverse Correlation with Education Level:** There is a clear trend where RSE increases as the level of education rises. For example, All-India Rural Male GAR_RSE grows from 0.70% (Primary) to 2.64% (Higher Secondary), reflecting smaller sample sizes in higher grades.
- **Statistical Instability in Small UTs:** Dadra & Nagar Haveli (State 26) exhibits extremely high RSE values, exceeding 100% in some Higher Secondary cohorts. These estimates should be considered statistically unreliable for independent analysis.
- **State-Level Reliability:** Larger states like Uttar Pradesh and Tamil Nadu maintain RSEs generally within the 2-11% range. While less precise than the national average, these values

are within acceptable limits for regional comparative studies.

- **Comparison of Metrics:** The RSE patterns for NAR and GAR are closely aligned, suggesting that the underlying sampling error affects both metrics similarly, regardless of the stricter age-criteria applied to NAR.
- **Urban vs. Rural Variance:** In several instances, Urban RSEs are slightly higher than Rural RSEs at the Secondary level, likely due to the higher diversity of educational paths (private, vocational, etc.) available in urban centers compared to rural areas.

Estimates of GAR by UMPCE quintile classes

Key Observations: GAR by UMPCE Quintile Classes

- **Economic Neutrality at Primary Level:** At the Primary level, GAR values are consistently near 100 across all quintiles (Q1–Q5). This suggests that basic education access is largely independent of household expenditure levels across the studied states.
- **Widening Wealth Gap in Higher Education:** The influence of household wealth becomes pronounced at higher levels. At the All-India Rural level for Higher Secondary, there is a 35-percentage-point gap between the poorest quintile (47.60) and the richest quintile (82.70).
- **Regional Success in Inclusion:** Tamil Nadu (State 33) exhibits high inclusivity; its Rural Secondary GAR for the lowest quintile (Q1) is 103.31, significantly outperforming the national Q1 average of 74.00.
- **Urban Bottom-Quintile Advantage:** Residents in the lowest urban quintile generally maintain higher attendance ratios than those in the lowest rural quintile. For Higher Secondary, the All-India Urban Q1 (59.64) is nearly 12 points higher than Rural Q1 (47.60).
- **Anomalies in Low-Population Units:** D & N Haveli (State 26) shows erratic quintile data, such as a GAR of 194.10 for Rural Q5 at the Higher Secondary level, which likely stems from a very small number of sample households in that specific expenditure bracket.
- **Saturation in Richer Households:** In Urban areas, the GAR for the top two quintiles (Q4 and Q5) frequently exceeds 90–100% across all course levels, indicating that economic stability is a primary driver for continued participation in the education system.

Table 5: GAR (RSE) by Sector, Course, and UMPCE Quintile

Sector	Course	Quintile	All India (-1)	UP (9)	D & N Haveli (26)	Tamil Nadu (33)
Rural	Primary	Q1	100.45 (1.20)	98.40 (2.84)	100.00 (0.00)	103.35 (9.04)
		Q2	100.14 (1.33)	99.29 (2.73)	88.54 (9.98)	100.08 (4.76)
		Q3	100.26 (1.06)	97.57 (2.95)	100.00 (0.00)	98.24 (3.39)
		Q4	101.87 (1.31)	104.35 (5.35)	100.00 (0.00)	104.82 (8.21)
		Q5	102.47 (1.33)	93.36 (6.13)	61.15 (55.41)	96.14 (4.40)
	Upper Primary	Q1	93.03 (2.14)	86.32 (4.43)	100.00 (0.00)	91.53 (9.43)
		Q2	91.96 (2.26)	87.36 (5.66)	117.42 (15.95)	94.29 (10.83)
		Q3	97.53 (2.01)	96.36 (5.31)	107.13 (8.73)	90.94 (8.09)
		Q4	96.22 (2.21)	92.61 (9.55)	77.42 (11.89)	95.46 (9.40)
		Q5	93.77 (1.98)	92.87 (8.63)	100.00 (0.00)	94.20 (4.87)
	Secondary	Q1	74.00 (3.01)	58.28 (7.30)	100.00 (0.00)	103.31 (12.60)
		Q2	79.33 (3.28)	60.02 (8.59)	86.01 (7.29)	98.60 (16.70)
		Q3	83.92 (3.43)	64.97 (10.00)	80.64 (24.30)	89.64 (16.03)
		Q4	91.10 (3.18)	61.12 (13.38)	116.84 (13.40)	88.29 (12.73)
		Q5	93.82 (2.61)	76.61 (12.62)	3.68 (187.04)	94.64 (7.05)
	Higher Sec.	Q1	47.60 (4.83)	46.45 (11.17)	0.00 (NaN)	72.60 (26.28)
		Q2	57.94 (4.45)	59.61 (10.48)	41.61 (71.65)	78.34 (21.21)
		Q3	62.45 (4.60)	65.27 (12.39)	0.00 (NaN)	98.28 (18.44)
		Q4	71.86 (3.93)	62.76 (11.99)	17.88 (106.55)	78.40 (11.63)
		Q5	82.70 (3.34)	71.59 (12.63)	194.10 (4.02)	93.98 (8.82)
Urban	Primary	Q1	102.32 (1.41)	99.85 (3.74)	100.85 (1.05)	121.62 (7.03)
		Q2	103.10 (1.49)	107.19 (5.01)	74.77 (25.07)	103.57 (4.90)
		Q3	100.97 (1.95)	89.31 (6.62)	118.50 (5.59)	112.75 (4.70)
		Q4	100.87 (1.88)	90.43 (6.89)	120.14 (33.53)	106.26 (7.93)
		Q5	103.78 (1.75)	105.00 (7.49)	122.47 (28.45)	95.11 (8.73)
	Upper Primary	Q1	87.18 (2.45)	68.94 (7.19)	79.38 (6.56)	86.26 (7.50)
		Q2	96.58 (2.86)	81.61 (9.11)	100.76 (1.97)	107.08 (8.05)
		Q3	95.07 (2.73)	118.64 (11.18)	76.64 (7.63)	89.80 (7.41)
		Q4	99.50 (2.64)	105.62 (13.62)	29.55 (138.12)	102.63 (4.99)
		Q5	94.32 (4.17)	96.36 (6.71)	81.65 (28.45)	112.40 (9.37)
	Secondary	Q1	80.27 (3.26)	61.94 (8.36)	282.62 (158.93)	88.28 (15.39)
		Q2	96.17 (3.89)	82.00 (13.20)	159.51 (19.60)	96.02 (11.37)
		Q3	95.67 (4.56)	70.92 (16.68)	127.52 (17.51)	77.33 (17.04)
		Q4	99.54 (3.77)	101.78 (17.10)	85.64 (23.90)	79.66 (12.51)
		Q5	101.85 (3.85)	88.37 (17.16)	100.00 (0.00)	82.59 (9.58)
	Higher Sec.	Q1	59.64 (4.71)	51.60 (9.32)	0.00 (NaN)	104.11 (17.05)
		Q2	71.27 (4.74)	60.63 (17.75)	22.58 (78.86)	79.34 (14.92)
		Q3	84.13 (5.56)	89.65 (24.09)	78.19 (20.42)	93.08 (16.62)
		Q4	91.05 (4.52)	87.04 (20.72)	2011.93 (45.94)	124.74 (16.34)
		Q5	99.31 (3.48)	108.36 (11.64)	162.19 (54.22)	94.09 (10.07)

Estimates of NAR by UMPCE quintile classes

Key Observations: NAR by UMPCE Quintile Classes

- **Direct Correlation with Wealth:** Net Attendance Ratio (NAR) shows a strong positive correlation with household expenditure across all levels. In Rural India, the Higher Secondary NAR for the richest quintile (53.33%) is nearly twice that of the poorest quintile (26.93%).

Table 6: NAR (RSE) by Sector, Course, and UMPCE Quintile

Sector	Course	Quintile	All India (-1)	UP (9)	D & N Haveli (26)	Tamil Nadu (33)
Rural	Primary	Q1	83.35 (1.14)	78.97 (2.75)	100.00 (0.00)	92.51 (6.46)
		Q2	85.43 (1.08)	82.23 (2.56)	88.54 (9.98)	92.43 (3.43)
		Q3	85.84 (0.88)	79.18 (2.60)	100.00 (0.00)	92.36 (3.20)
		Q4	87.74 (0.90)	79.20 (4.00)	100.00 (0.00)	83.98 (4.81)
		Q5	88.00 (1.03)	76.40 (5.98)	61.15 (55.41)	87.14 (3.42)
	Upper Primary	Q1	66.48 (1.88)	55.04 (5.11)	100.00 (0.00)	84.81 (8.24)
		Q2	69.02 (2.13)	61.29 (5.40)	100.00 (0.00)	78.53 (8.89)
		Q3	72.97 (1.95)	63.11 (5.16)	100.00 (0.00)	76.86 (8.24)
		Q4	73.47 (1.85)	57.10 (8.19)	77.42 (11.89)	73.26 (9.49)
		Q5	77.36 (1.81)	67.95 (9.07)	100.00 (0.00)	80.19 (4.44)
	Secondary	Q1	47.15 (3.51)	32.55 (9.93)	100.00 (0.00)	84.90 (10.75)
		Q2	54.07 (3.21)	42.27 (9.07)	86.01 (7.29)	66.31 (14.79)
		Q3	54.81 (3.23)	39.64 (10.37)	77.09 (26.44)	61.75 (12.89)
		Q4	59.46 (3.08)	29.51 (14.52)	73.37 (17.74)	59.87 (12.46)
		Q5	67.08 (2.70)	51.76 (14.68)	3.45 (193.10)	71.91 (6.88)
	Higher Sec.	Q1	26.93 (5.67)	24.97 (13.45)	0.00 (NaN)	60.82 (26.24)
		Q2	36.52 (4.72)	35.84 (10.76)	41.61 (71.65)	46.23 (21.13)
		Q3	38.22 (4.46)	30.70 (11.70)	0.00 (NaN)	63.63 (15.69)
		Q4	46.20 (3.96)	32.88 (12.13)	17.88 (106.55)	58.41 (14.76)
		Q5	53.33 (3.39)	37.79 (15.15)	99.59 (0.48)	67.45 (7.80)
Urban	Primary	Q1	83.90 (1.09)	72.70 (3.04)	100.00 (0.00)	97.49 (1.78)
		Q2	86.92 (1.19)	80.34 (2.79)	74.26 (24.36)	86.32 (5.88)
		Q3	86.92 (1.43)	80.43 (6.38)	100.00 (0.00)	95.91 (1.80)
		Q4	88.58 (1.37)	76.67 (5.61)	100.00 (0.00)	87.86 (4.20)
		Q5	91.97 (1.22)	88.73 (4.77)	100.00 (0.00)	86.59 (6.20)
	Upper Primary	Q1	66.53 (2.22)	47.61 (7.55)	79.38 (6.56)	81.71 (6.87)
		Q2	75.93 (2.48)	60.66 (10.76)	98.85 (1.34)	85.49 (4.92)
		Q3	75.97 (2.22)	82.70 (5.72)	76.64 (7.63)	80.00 (6.24)
		Q4	79.30 (2.15)	64.59 (13.35)	15.37 (134.57)	89.37 (4.35)
		Q5	76.86 (3.61)	86.00 (8.42)	81.65 (28.45)	89.59 (4.64)
	Secondary	Q1	52.99 (3.82)	35.03 (9.88)	14.21 (19.59)	65.70 (19.78)
		Q2	60.00 (3.52)	48.63 (10.16)	97.55 (3.21)	70.42 (11.61)
		Q3	63.86 (4.25)	47.29 (18.07)	100.00 (0.00)	48.77 (15.86)
		Q4	69.53 (3.54)	57.30 (16.56)	14.36 (142.57)	66.87 (14.78)
		Q5	70.95 (3.80)	61.95 (22.59)	100.00 (0.00)	70.64 (9.70)
	Higher Sec.	Q1	38.18 (5.23)	33.03 (11.99)	0.00 (NaN)	63.79 (15.45)
		Q2	45.82 (5.14)	34.33 (17.84)	8.87 (17.75)	54.48 (14.16)
		Q3	51.75 (4.70)	40.32 (18.28)	78.19 (20.42)	52.48 (13.66)
		Q4	58.98 (4.39)	51.46 (16.15)	67.52 (45.94)	70.31 (8.97)
		Q5	73.86 (3.75)	78.26 (11.44)	100.00 (0.00)	72.11 (9.51)

- **Age-Appropriate Gaps at Primary Level:** Even at the Primary level, where access is near-universal, NAR varies by wealth. In Urban areas, the Q5 quintile NAR (91.97%) is significantly higher than the Q1 NAR (83.90%), indicating that children from poorer backgrounds are more likely to be over-age or under-age for their grade.
- **Social Equity in Tamil Nadu:** Tamil Nadu (State 33) demonstrates high educational efficiency; its Rural Secondary NAR for the poorest quintile (84.90%) is remarkably higher than the All-India average for the wealthiest quintile (67.08%).
- **Vulnerability in Rural Uttar Pradesh:** Uttar Pradesh (State 9) exhibits a severe drop in age-appropriate attendance for the lower quintiles at the Secondary level (32.55% for Rural Q1), highlighting a critical need for targeted interventions in poor rural areas.

- **The Urban-Rural Quintile Gap:** For the same economic bracket (Q1), Urban students maintain a consistently higher NAR than Rural students. At the Higher Secondary level, Urban Q1 (38.18%) outperforms Rural Q1 (26.93%) by a substantial margin.
- **Impact of Economic Barriers:** The steep decline in NAR from Primary to Higher Secondary for Q1 households (83.35% to 26.93% at All-India Rural) reflects the high opportunity cost of education for low-income families as children reach working age.

Difference between GAR and NAR across gender & sector

Key Observations: Difference Between GAR and NAR

- **Accumulation of Age-Grade Mismatch:** There is a distinct trend where the ($GAR - NAR$) gap widens as students progress. Nationally, the gap increases from roughly 15 percentage points at the Primary level to nearly 30 points at the Secondary level, highlighting the cumulative effect of late starts and grade repetitions.
- **High Distortion in Uttar Pradesh:** State 9 (UP) exhibits significant age-grade distortion, particularly in Rural Upper Primary education (gap of 31.62% for females). This reflects a system where a third of the attending students are outside the official age bracket for their grade.
- **Persistence in Tamil Nadu:** In Tamil Nadu (State 33), the Rural Higher Secondary female gap (35.63%) is one of the highest observed. This suggests a high level of persistence, where students remain in the education system to complete their schooling despite being over-age.
- **Urban Secondary Peaks:** For Urban areas at the All-India level, the age-grade gap peaks at the Secondary level (32.25% for males), suggesting that the transition from middle to high school in urban centers involves significant age-variance.

Table 7: Difference Between Gross and Net Attendance Ratios ($GAR - NAR$)

Sector	Course	Gender	All India	UP (9)	D & N Haveli (26)	Tamil Nadu (33)
Rural	Primary	Male	15.15	20.81	0.00	10.22
		Female	15.08	17.06	0.00	11.16
	Upper Primary	Male	22.68	29.69	4.74	14.72
		Female	23.42	31.62	8.94	16.27
	Secondary	Male	28.62	27.65	5.46	21.82
		Female	27.17	20.59	5.26	29.13
	Higher Sec.	Male	26.09	32.34	1.00	19.83
		Female	21.89	22.15	105.76	35.63
Urban	Primary	Male	14.73	26.13	19.66	12.25
		Female	15.74	18.43	0.00	25.95
	Upper Primary	Male	20.78	22.45	0.00	23.59
		Female	18.78	24.48	5.98	4.12
	Secondary	Male	32.25	33.66	40.55	23.26
		Female	30.02	24.79	54.67	16.49
	Higher Sec.	Male	27.04	33.00	61.49	34.16
		Female	26.81	20.21	0.78	37.55

- **Gendered Persistence Trends:** In rural Uttar Pradesh, the gap for males (32.34%) exceeds that for females (22.15%) at the Higher Secondary level, indicating that over-age male students are more likely to remain enrolled than their female counterparts.
- **Outliers in Small Jurisdictions:** The extreme gap of 105.76% recorded for Rural Females in D & N Haveli (State 26) is a statistical outlier, likely resulting from the small sample size and high variance previously identified in the RSE analysis.

Necessary R-codes

1. Data Collection

```

1 library(readr)
2 library(haven)
3
4 data_3 = read_sav(
5   "Block3.sav")
6 colnames(data_3) <- c(
7   "hhid", "centre_round", "fsu", "round", "schedule",
8   "sample", "sector", "nss_region", "state", "district",
9   "state_district", "stratum", "sub_stratum", "sub_round",
10  "sub_sample", "fod_sub_region", "hamlet_sub_block",
11  "second_stage_stratum", "sample_hhid", "level", "filler",
12  "household_size", "nic_2008_code", "nco_2004_3d_code",
13  "household_type", "hh_type", "religion", "social_group",
14  "distance_primary_school", "distance_upper_school",
15  "distance_secondary_school", "hh_computer", "member_internet",
16  "any_hh_member_3_35_attending", "hostel",
17  "location_state", "location_district", "location_sector",
18  "hh_cons_exp_rs", "nss", "nsc", "mult",
19  "mult_subsample", "mult_combined")
20
21 data_4 <- read_tsv("Block-4.txt")
22 colnames(data_4) <- c(
23   "hhid", "centre_round", "fsu", "round", "schedule",
24   "sample", "sector", "nss_region", "state", "district",
25   "state_district", "stratum", "sub_stratum", "sub_round",
26   "sub_sample", "fod_sub_region", "hamlet_sub_block",
27   "second_stage_stratum", "sample_hhid", "level", "filler",
28   "person_serial_no", "relation_to_head", "gender", "age",
29   "marital_status", "edu_level_general", "edu_level_technical",
30   "completed_class_grade", "edu_completed_in_yrs",
31   "current_edu_status_3_35yrs", "operate_computer_5yrs",
32   "operate_internet_5yrs", "internet_use_last30days_5yrs",
33   "prev_voc_tech_training_3_35yrs", "disability_certificate",
34   "disability_type", "nss", "nsc", "mult",
35   "mult_subsample", "mult_combined")
36
37 data_5 <- read_tsv("Block-5.txt")
38 colnames(data_5) <- c(
39   "hhid", "centre_round", "fsu", "round", "schedule",
40   "sample", "sector", "nss_region", "state", "district",
41   "state_district", "stratum", "sub_stratum", "sub_round",
42   "sub_sample", "fod_sub_region", "hamlet_sub_block",
43   "second_stage_stratum", "sample_hhid", "level", "filler",
44   "person_serial_no", "age", "school_entry_age",
45   "language_spoken_home", "medium_instruction",
46   "enrol_basic_course", "course_attending", "type_course",
47   "course_duration_months", "same_class_as_previous_year",
48   "institution_type", "nature_institution",
49   "reason_private_inst1", "reason_private_inst2",

```

```

50 "attend_special_school_disability", "free_education",
51 "tuition_fee_payable", "tuition_fee_reason_not_pay",
52 "tuition_fee_waived_amount", "scholarship_stipend_received",
53 "scholarship_stipend_amount", "reimbursement_type",
54 "reimbursement_agency", "free_textbooks", "free_stationery",
55 "midday_meal_tiffin", "midday_meal_agency", "transport_mode",
56 "concession_received", "distance_institution",
57 "changed_institution_365days", "taking_private_coaching",
58 "nss", "nsc", "mult", "mult_subsample", "mult_combined")

```

2. Data Preparation and Merging

```

1 library(Hmisc)
2
3 # Compute Usual Monthly Per Capita Expenditure (UMPCE)
4 data_3$umpce <- as.numeric(data_3$hh_cons_exp_rs) /
5                 as.numeric(data_3$household_size)
6
7 # Assign UMPCE quintiles separately for Rural (1) and Urban (2)
8 data_3$umpce_quintile <- NA_character_
9
10 for (s in c("1", "2")) {
11   idx <- data_3$sector == s
12   q <- wtd.quantile(
13     data_3$umpce[idx],
14     weights = as.numeric(data_3$household_size[idx]) *
15              as.numeric(data_3$mult_combined[idx]),
16     probs = c(0.2, 0.4, 0.6, 0.8)
17   )
18   data_3$umpce_quintile[idx] <- paste0("Q",
19     as.numeric(cut(
20       data_3$umpce[idx],
21       breaks = c(-Inf, q, Inf),
22       include.lowest = TRUE
23     ))
24   )
25 }
26
27 data_3_red <- data_3[, c("hhid", "umpce_quintile")]
28
29 # -- Block 4: age groups and gender / sector labels -----
30 data_4 <- data_4[!is.na(data_4$age), ]
31 data_4$age_group <- cut(
32   as.numeric(data_4$age),
33   breaks = c(-Inf, 5, 10, 13, 15, 17, Inf),
34   labels = c("5-", "6-10", "11-13", "14-15", "16-17", "18+"),
35   right = TRUE, include.lowest = TRUE
36 )
37 data_4$gender[data_4$gender == 1] <- "Male"
38 data_4$gender[data_4$gender == 2] <- "Female"
39 data_4$gender[data_4$gender == 3] <- "Others"
40 data_4$sector[data_4$sector == 1] <- "Rural"
41 data_4$sector[data_4$sector == 2] <- "Urban"
42
43 data_4_red <- data_4[, c(
44   "hhid", "stratum", "sub_stratum", "sub_sample",
45   "sector", "state", "person_serial_no", "gender",
46   "age_group", "mult", "mult_subsample", "mult_combined")]
47
48 # -- Block 5: course level groups -----
49 data_5$course_group <- "Others"
50 data_5$course_group[data_5$enrol_basic_course == "07"] <- "Primary"

```

```

51 data_5$course_group[data_5$enrol_basic_course == "08"] <-
52   "Upper Primary / Middle"
53 data_5$course_group[data_5$enrol_basic_course == "10"] <- "Secondary"
54 data_5$course_group[data_5$enrol_basic_course == "11"] <-
55   "Higher Secondary"
56
57 data_5_red <- data_5[, c("hhid", "person_serial_no", "course_group")]
58
59 # -- Merge Blocks 4, 5, and 3 -----
60 merged <- merge(data_4_red, data_5_red,
61                 by = c("hhid", "person_serial_no"), all.x = TRUE)
62 merged$course_group[is.na(merged$course_group)] <- "Not Applicable"
63 merged <- merge(merged, data_3_red, by = "hhid", all.x = TRUE)

```

3. Estimation Functions

```

1  # Internal helper: RSE (%) for a ratio R = Y/X via interpenetrating
2  # sub-samples
3  .mse_ratio <- function(R, X, y_wt, x_wt,
4                        stratum, sub_stratum, sub_sample) {
5    tmp <- data.frame(stratum, sub_stratum, sub_sample,
6                      Y = y_wt, X = x_wt)
7    agg <- aggregate(cbind(Y, X) ~
8                     stratum + sub_stratum + sub_sample,
9                     data = tmp, sum)
10   st1 <- agg[agg$sub_sample == 1, ]
11   st2 <- agg[agg$sub_sample == 2, ]
12   dY <- st1$Y - st2$Y
13   dX <- st1$X - st2$X
14   MSE <- sum(dY^2 + R^2 * dX^2 - 2 * R * dY * dX) / (4 * X^2)
15   sqrt(MSE) * 100 / R # RSE in percent
16 }
17
18 # Net Attendance Rate (NAR) - by Gender
19 NAR <- function(course_group, age_group, gender, sector, state) {
20   base <- merged$gender == gender & merged$sector == sector
21   if (state != -1)
22     base <- base & as.numeric(merged$state) == state
23
24   y <- as.numeric(base &
25                   merged$course_group == course_group &
26                   merged$age_group == age_group)
27   x <- as.numeric(base & merged$age_group == age_group)
28
29   Y <- sum(merged$mult_combined * y)
30   X <- sum(merged$mult_combined * x)
31   R <- Y / X
32   RSE <- .mse_ratio(R, X,
33                     merged$mult_subsample * y,
34                     merged$mult_subsample * x,
35                     merged$stratum, merged$sub_stratum, merged$sub_sample)
36   c(R, RSE)
37 }
38
39 # Gross Attendance Rate (GAR) - by Gender
40 GAR <- function(course_group, age_group, gender, sector, state) {
41   base <- merged$gender == gender & merged$sector == sector
42   if (state != -1)
43     base <- base & as.numeric(merged$state) == state
44
45   y <- as.numeric(base & merged$course_group == course_group)
46   x <- as.numeric(base & merged$age_group == age_group)

```

```

47
48 Y <- sum(merged$mult_combined * y)
49 X <- sum(merged$mult_combined * x)
50 R <- Y / X
51 RSE <- .mse_ratio(R, X,
52                 merged$mult_subsample * y,
53                 merged$mult_subsample * x,
54                 merged$stratum, merged$sub_stratum, merged$sub_sample)
55 c(R, RSE)
56 }
57
58 # Net Attendance Rate (NAR) - by UMPCE Quintile
59 UMPCE_NAR <- function(course_group, age_group,
60                       umpce_quintile, sector, state) {
61   base <- merged$umpce_quintile == umpce_quintile &
62         merged$sector == sector
63   if (state != -1)
64     base <- base & as.numeric(merged$state) == state
65
66   y <- as.numeric(base &
67                 merged$course_group == course_group &
68                 merged$age_group == age_group)
69   x <- as.numeric(base & merged$age_group == age_group)
70
71   Y <- sum(merged$mult_combined * y)
72   X <- sum(merged$mult_combined * x)
73   R <- Y / X
74   RSE <- .mse_ratio(R, X,
75                   merged$mult_subsample * y,
76                   merged$mult_subsample * x,
77                   merged$stratum, merged$sub_stratum, merged$sub_sample)
78   c(R, RSE)
79 }
80
81 # Gross Attendance Rate (GAR) - by UMPCE Quintile
82 UMPCE_GAR <- function(course_group, age_group,
83                       umpce_quintile, sector, state) {
84   base <- merged$umpce_quintile == umpce_quintile &
85         merged$sector == sector
86   if (state != -1)
87     base <- base & as.numeric(merged$state) == state
88
89   y <- as.numeric(base & merged$course_group == course_group)
90   x <- as.numeric(base & merged$age_group == age_group)
91
92   Y <- sum(merged$mult_combined * y)
93   X <- sum(merged$mult_combined * x)
94   R <- Y / X
95   RSE <- .mse_ratio(R, X,
96                   merged$mult_subsample * y,
97                   merged$mult_subsample * x,
98                   merged$stratum, merged$sub_stratum, merged$sub_sample)
99   c(R, RSE)
100 }

```

4. Computing GAR and NAR by Gender

```

1 course_age_pairs <- data.frame(
2   course      = c("Primary", "Upper Primary / Middle",
3                 "Secondary", "Higher Secondary"),
4   age_group   = c("6-10", "11-13", "14-15", "16-17"),
5   stringsAsFactors = FALSE

```



```

6 )
7
8 other_params <- expand.grid(
9   gender = c("Male", "Female"),
10  sector = c("Rural", "Urban"),
11  state = c(-1, 9, 26, 33),
12  stringsAsFactors = FALSE
13 )
14
15 params <- merge(course_age_pairs, other_params, by = NULL)
16 params[, c("GAR", "GAR_RSE", "NAR", "NAR_RSE")] <- NA
17
18 pb <- txtProgressBar(min = 0, max = nrow(params), style = 3)
19
20 for (i in seq_len(nrow(params))) {
21   gar <- GAR(params$course[i], params$age_group[i],
22             params$gender[i], params$sector[i], params$state[i])
23   nar <- NAR(params$course[i], params$age_group[i],
24             params$gender[i], params$sector[i], params$state[i])
25   params$GAR[i] <- gar[1]; params$GAR_RSE[i] <- gar[2]
26   params$NAR[i] <- nar[1]; params$NAR_RSE[i] <- nar[2]
27   setTxtProgressBar(pb, i)
28 }
29 close(pb)

```

5. Computing GAR and NAR by UMPCE Quintile

```

1 other_params2 <- expand.grid(
2   umpce_quintile = paste0("Q", 1:5),
3   sector = c("Rural", "Urban"),
4   state = c(-1, 9, 26, 33),
5   stringsAsFactors = FALSE
6 )
7
8 params2 <- merge(course_age_pairs, other_params2, by = NULL)
9 params2[, c("GAR", "GAR_RSE", "NAR", "NAR_RSE")] <- NA
10
11 pb <- txtProgressBar(min = 0, max = nrow(params2), style = 3)
12
13 for (i in seq_len(nrow(params2))) {
14   gar <- UMPCE_GAR(params2$course[i], params2$age_group[i],
15                   params2$umpce_quintile[i],
16                   params2$sector[i], params2$state[i])
17   nar <- UMPCE_NAR(params2$course[i], params2$age_group[i],
18                   params2$umpce_quintile[i],
19                   params2$sector[i], params2$state[i])
20   params2$GAR[i] <- gar[1]; params2$GAR_RSE[i] <- gar[2]
21   params2$NAR[i] <- nar[1]; params2$NAR_RSE[i] <- nar[2]
22   setTxtProgressBar(pb, i)
23 }
24 close(pb)

```

6. Exporting Results

```

1 write.csv(as.data.frame(params), "result1.csv", row.names = FALSE)
2 write.csv(as.data.frame(params2), "result2.csv", row.names = FALSE)

```